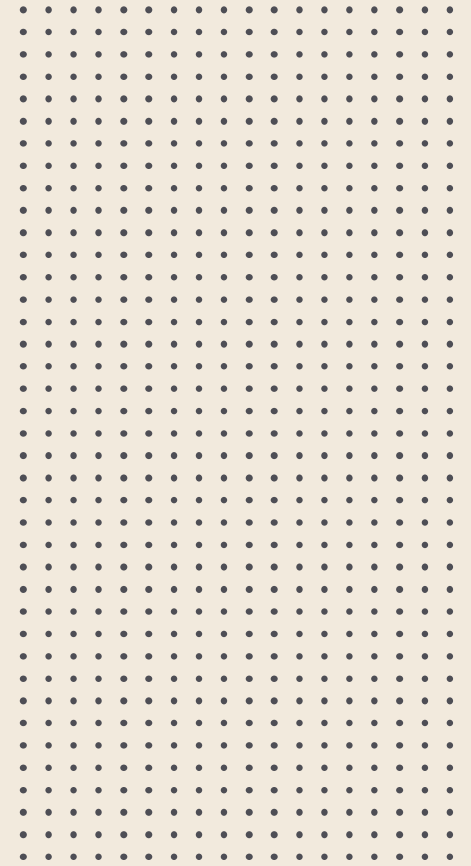


CSE-1223

Discrete Mathematics

Segment 4: Number Theory

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Segment Motivation

- **Number theory** is the part of mathematics devoted to the study of the integers and their properties.
- Key ideas in number theory include divisibility and the primality of integers.
- Representations of integers, including binary and hexadecimal representations, are part of number theory.
- Number theory has long been studied because of the beauty of its ideas, its accessibility, and its wealth of open questions.
- We'll use many ideas developed in Chapter 1 about proof methods and proof strategy in our exploration of number theory.

Division

DEFINITION 1

If a and b are integers with $a \neq 0$, we say that a *divides* b if there is an integer c such that $b = ac$, or equivalently, if $\frac{b}{a}$ is an integer. When a divides b we say that a is a *factor* or *divisor* of b , and that b is a *multiple* of a . The notation $a \mid b$ denotes that a divides b . We write $a \nmid b$ when a does not divide b .

EXAMPLE 1 Determine whether $3 \mid 7$ and whether $3 \mid 12$.

Solution: We see that $3 \nmid 7$, because $7/3$ is not an integer. On the other hand, $3 \mid 12$ because $12/3 = 4$. 

Division

EXAMPLE 2 Let n and d be positive integers. How many positive integers not exceeding n are divisible by d ?

Solution: The positive integers divisible by d are all the integers of the form dk , where k is a positive integer. Hence, the number of positive integers divisible by d that do not exceed n equals the number of integers k with $0 < dk \leq n$, or with $0 < k \leq n/d$. Therefore, there are $\lfloor n/d \rfloor$ positive integers not exceeding n that are divisible by d . ◀



Division

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Let $n=15$, $d = 3$

$3 \cdot 1 = 3$ $3 \cdot 2 = 6$ $3 \cdot 3 = 9$ $3 \cdot 4 = 12$ $3 \cdot 5 = 15$

Division

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Let $n=15$, $d=3$

$3 \cdot 1 = 3$ $3 \cdot 2 = 6$ $3 \cdot 3 = 9$ $3 \cdot 4 = 12$ $3 \cdot 5 = 15$

Let $n=21$, $d=7$

$7 \cdot 1 = 7$ $7 \cdot 2 = 14$ $7 \cdot 3 = 21$

Division

THEOREM 1

Let a , b , and c be integers, where $a \neq 0$. Then

- (i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$; Aut'16
- (ii) if $a \mid b$, then $a \mid bc$ for all integers c ;
- (iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

Proof:

i) Suppose that $a \mid b$ and $a \mid c$. Then, from the definition of divisibility, it follows that there are integers s and t with $b = as$ and $c = at$. Hence,
 $b + c = as + at = a(s + t)$.

Therefore, a divides $b + c$. This establishes part (i) of the theorem.

Home Work: 1. Prove that part (ii) of Theorem 1 is true. 2. Prove that part (iii) of Theorem 1 is true.
3. If a , b , and c are integers, where $a \neq 0$, such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.



Division Algorithm

- When an integer is divided by a positive integer, there is a quotient and a remainder. The statement below is traditionally called the “Division Algorithm,” but is really a theorem.

Theorem 2: Division Algorithm: If a is an integer and d a positive integer, then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$

- d is called the divisor.
- a is called the dividend.
- q is called the quotient.
- r is called the remainder.

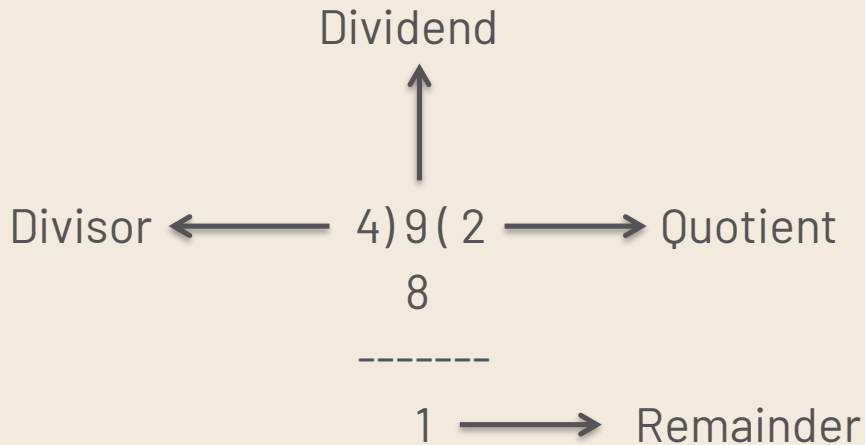
Division Algorithm

DEFINITION 2

In the equality given in the division algorithm, d is called the *divisor*, a is called the *dividend*, q is called the *quotient*, and r is called the *remainder*. This notation is used to express the quotient and remainder:

$$q = a \text{ \textbf{div} } d, \quad r = a \text{ \textbf{mod} } d.$$

Division Algorithm



Examples:

- What are the quotient and remainder when 101 is divided by 11?
- What are the quotient and remainder when -11 is divided by 3?



Division Algorithm



Examples:

- What are the quotient and remainder when 101 is divided by 11?

Solution:

The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.



Division Algorithm

Examples:

- What are the quotient and remainder when 101 is divided by 11?

Solution:

The quotient when 101 is divided by 11 is $9 = 101 \text{ div } 11$, and the remainder is $2 = 101 \text{ mod } 11$.

- What are the quotient and remainder when -11 is divided by 3?

Solution:

The quotient when -11 is divided by 3 is $-4 = -11 \text{ div } 3$, and the remainder is $1 = -11 \text{ mod } 3$.

Definitions of Functions div and mod



Modular Arithmetic



Definition 3: If a and b are integers and m is a positive integer, then **a is congruent to b modulo m** if m divides $a - b$.

- The notation $a \equiv b \pmod{m}$ says that a is congruent to b modulo m .
- We say that $a \equiv b \pmod{m}$ is a congruence and that m is its modulus.
- Two integers are congruent mod m if and only if they have the same remainder when divided by m .
- If a is not congruent to b modulo m , we write
$$a \not\equiv b \pmod{m}$$

Theorem 3: Let a and b be integers, and let m be a positive integer. Then $a \equiv b \pmod{m}$ if and only if $a \bmod m = b \bmod m$.



Modular Arithmetic

Applications:

- Modular arithmetic is the basis of computing.
- Used with two's complement representation to implement computer arithmetic.
- Also used in hashing, pseudo-random number generation, and cryptography.
- It's often used to detect errors in identification numbers.
- It is used to calculate checksums for ISBNs and Ibans.



Modular Arithmetic



Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.



Modular Arithmetic

Example: Determine whether 17 is congruent to 5 modulo 6 and whether 24 and 14 are congruent modulo 6.

Solution:

$17 \equiv 5 \pmod{6}$ because 6 divides $17 - 5 = 12$.

$24 \not\equiv 14 \pmod{6}$ since 6 divides $24 - 14 = 10$ is not divisible by 6.



Modular Arithmetic



Theorem 4: Let m be a positive integer. The integers a and b are congruent modulo m if and only if there is an integer k such that $a = b + km$.

Proof:

If $a \equiv b \pmod{m}$, then (by the definition of congruence) $m \mid a - b$. Hence, there is an integer k such that $a - b = km$ and equivalently $a = b + km$.

Conversely, if there is an integer k such that $a = b + km$, then $km = a - b$. Hence, m divides $a - b$, so that $a \equiv b \pmod{m}$.



Modular Arithmetic



Theorem 5: Let m be a positive integer. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

Proof:

Because $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, by Theorem 4 there are integers s and t with $b = a + sm$ and $d = c + tm$.

Therefore,

$$b + d = (a + sm) + (c + tm) = (a + c) + m(s + t)$$

$$bd = (a + sm)(c + tm) = ac + m(at + cs + stm).$$

Hence, $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$.

Modular Arithmetic

Example: IF $7 \equiv 2 \pmod{5}$ and $11 \equiv 1 \pmod{5}$, show that $18 \equiv 3 \pmod{5}$ and $77 \equiv 2 \pmod{5}$

Solution:

From theorem 5, we know that,

If m be a positive integer, $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $a + c \equiv b + d \pmod{m}$ and $ac \equiv bd \pmod{m}$

So, $18 = 7 + 11 \equiv 2 + 1 = 3 \pmod{5}$

And $77 = 7 \times 11 \equiv 2 \times 1 = 2 \pmod{5}$



Modular Arithmetic

Theorem 6:

Let m be a positive integer and let a and b be integers. Then

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

and

$$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$$

Modular Arithmetic

Example: Find each of these values.

a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

Solution:

a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

We know,

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m$$

According to the Theorem, we have

$$(-133 + 261) \bmod 23$$

$$= 128 \bmod 23$$

$$= 13$$

$$\begin{array}{r} 23 \overline{) 128} \end{array} \begin{array}{l} 5 \\ 115 \\ \hline 13 \end{array}$$

Modular Arithmetic

Example: Find each of these values.

a) $(-133 \bmod 23 + 261 \bmod 23) \bmod 23$

b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

Solution:

b) $(457 \bmod 23 \cdot 182 \bmod 23) \bmod 23$

We know,

$ab \bmod m = ((a \bmod m)(b \bmod m)) \bmod m.$

According to the Theorem, we have

$(457 \cdot 182) \bmod 23$

$= 83174 \bmod 23$

$= 6$

$23 \overline{) 83174} \quad (3616$

83168

6



Prime Number

Definition: An integer p greater than 1 is called **prime** if the only positive factors of p are 1 and p . A positive integer that is greater than 1 and is not prime is called **composite**. Aut'17

Example: The integer 7 is prime because its only positive factors are 1 and 7, but 9 is composite because it is divisible by 3.



Prime Number



Theorem 1: The Fundamental Theorem of Arithmetic

Theorem: Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.

Example: The prime factorizations of 100, 641, 999, and 1024 are given by

$$100 = 2 \cdot 2 \cdot 5 \cdot 5 = 2^2 \cdot 5^2$$

$$641 = 641$$

$$999 = 3 \cdot 3 \cdot 3 \cdot 37 = 3^3 \cdot 37$$

$$1024 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^{10}$$

Prime Number

Theorem 2: If n is a composite integer, then n has a prime divisor less than or equal to $\sqrt{n}$. Spr'18

Proof: If n is composite, by the definition of a composite integer, we know that it has a factor a with $1 < a < n$. Hence, by the definition of a factor of a positive integer, we have $n = ab$, where b is a positive integer greater than 1. We will show that $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. If $a > \sqrt{n}$ and $b > \sqrt{n}$, then $ab > \sqrt{n} \cdot \sqrt{n} = n$, which is a contradiction. Consequently, $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$. Because both a and b are divisors of n , we see that n has a positive divisor not exceeding $\sqrt{n}$. This divisor is either prime or, by the fundamental theorem of arithmetic, has a prime divisor less than itself. In either case, n has a prime divisor less than or equal to $\sqrt{n}$. ◀

Example: Show that 101 is prime.

Solution: The only primes not exceeding $\sqrt{101}$ are 2, 3, 5, and 7. Because 101 is not divisible by 2, 3, 5, or 7 (the quotient of 101 and each of these integers is not an integer), it follows that 101 is prime.



Prime Number

Define Mersenne Prime with example. Spr'16, Aut'17

Answer: Mersenne Prime is a prime number that can be written in the form $2^n - 1$. For example 31 is a Mersenne prime that can be written as $2^5 - 1$. The first few Mersenne primes are 3, 7, 31, 127, 8191. There are 50 known Mersenne primes as of June 2018, though we hope that it will change in the future.

Greatest Common Divisors

Define Greatest Common Divisors.

Definition: Let a and b be integers, not both zero. The largest integer d such that $d \mid a$ and $d \mid b$ is called the greatest common divisor of a and b . The greatest common divisor of a and b is denoted by $\gcd(a, b)$.

What is the greatest common divisor of 24 and 36?

Solution: The positive common divisors of 24 and 36 are 1, 2, 3, 4, 6, and 12. Hence, $\gcd(24, 36) = 12$.

What is the greatest common divisor of 17 and 22?

Solution: The integers 17 and 22 have no positive common divisors other than 1, so that $\gcd(17, 22) = 1$.

Greatest Common Divisors

Example: Find the GCD/GCF/HCF of 120 and 500 using Prime Factorization. Aut'16

Solution:

Steps to find GCD

1. Find the prime factorization of 120

$$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3^1 \times 5^1$$

2. Find the prime factorization of 500

$$500 = 2 \times 2 \times 5 \times 5 \times 5 = 2^2 \times 5^3$$

3. To find the GCD, multiply all the prime factors common to both numbers:

$$\text{Therefore, GCD} = 2^2 \times 5^1 = 20$$

Home Work: 4. Find the highest common factor of 34, 42 and 58. 5. Find the prime factorization of 118724

Spr'23



Greatest Common Divisors

Define relative prime.

Spr'16, Aut'16

Definition: The integers a and b are relatively prime if their greatest common divisor is 1.

Example: Show that 17 and 22 are relatively prime.

Solution: The integers 17 and 22 have no positive common divisors other than 1, so that $\gcd(17, 22) = 1$. So, 17 and 22 are relatively prime.



Least Common Multiple



Define Least Common Multiple (LCM).

Definition: The least common multiple of the positive integers a and b is the smallest positive integer that is divisible by both a and b . The least common multiple of a and b is denoted by $\text{lcm}(a, b)$.

Least Common Multiple

Example: Find the least common multiple (LCM) of 60 and 90

Solution: Let us find the LCM of 60 and 90 using the prime factorization method.

The prime factorization of 60 and 90 are:

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3^1 \times 5^1$$

and

$$90 = 2 \times 3 \times 3 \times 5 = 2^1 \times 3^2 \times 5^1$$

Now, we will find the product of only those factors that have the highest powers among these. This will be, $2^2 \times 3^2 \times 5^1 = 4 \times 9 \times 5 = 180$

Home Work: 6. Find $\gcd(1000, 625)$ and $\text{lcm}(1000, 625)$ and verify that $\gcd(1000, 625) * \text{lcm}(1000, 625) = 1000 * 625$

Spr'Aut'17

Euclidean Algorithm

Write the Euclidean Algorithm to find the GCD. Spr'18

Solution:

```
procedure gcd(a, b: positive integers)
  x := a
  y := b
  while y ≠ 0
    r := x mod y
    x := y
    y := r
  return x{gcd(a, b) is x}
```

Practical Problem:

- 10104 - Euclid Problem
- GYM - (J) Once Upon A Time
- UVA - 12775 - Gift Dilemma

Euclidean Algorithm

Find the greatest common divisor of 414 and 662 using the Euclidean algorithm.

Solution: Successive uses of the division algorithm give:

$$662 = 414 \cdot 1 + 248$$

$$414 = 248 \cdot 1 + 166$$

$$248 = 166 \cdot 1 + 82$$

$$166 = 82 \cdot 2 + 2$$

$$82 = 2 \cdot 41.$$

Hence, $\gcd(414, 662) = 2$, because 2 is the last nonzero remainder.





Extended Euclidean Algorithm



Extended Euclidean Algorithm is an extension of Euclidean Algorithm which finds two things for integer a and b :

1. It finds the value of $\gcd(a,b)$.
2. It finds two integers x and y such that, $ax+by=\gcd(a,b)$.

The expression $ax+by = \gcd(a,b)$ is known as Bezout's identity and the pair (x,y) that satisfies the identity is called Bezout coefficients.

Extended Euclidean Algorithm

Example: Express $\gcd(252, 198) = 18$ as a linear combination of 252 and 198. **Spr'18'23**

Solution: To show that $\gcd(252, 198) = 18$, the Euclidean algorithm uses these divisions:

$$252 = 1 \cdot 198 + 54$$

$$198 = 3 \cdot 54 + 36$$

$$54 = 1 \cdot 36 + 18$$

$$36 = 2 \cdot 18$$

Now we can write as

$$\begin{aligned} 18 &= 54 - 1 \cdot 36 \\ &= 54 - 1 \cdot (198 - 3 \cdot 54) \\ &= 54 - 198 + 3 \cdot 54 \\ &= 4 \cdot 54 - 198 \\ &= 4 \cdot (252 - 1 \cdot 198) - 198 \\ &= 4 \cdot 252 - 4 \cdot 198 - 198 \\ &= 4 \cdot 252 - 5 \cdot 198 \end{aligned}$$

Home Work: 7. Use the extended Euclidean algorithm to express $\gcd(26, 91)$ as a linear combination of 26 and 91. **Aut'17**

Inverse Modulo

From Extended Euclidean Algorithm, we know that

$$ax+by=\gcd(a,b)$$

If $\gcd(a,b) = 1$ then we have, $ax+by = 1$

Here, y is an inverse of a modulo b .

Example: Find an inverse of 3 modulo 7 Spr'17

Solution: $7 = 2 \cdot 3 + 1$

$$\Rightarrow 1 = 7 - 2 \cdot 3$$

$$\Rightarrow 1 = -2 \cdot 3 + 1 \cdot 7$$

Comparing with the above equation, -2 is an inverse of 3 modulo 7.

Home Work: 8. Find an inverse of 3 modulo 7 Aut'17

9. Find an inverse of 5 modulo 11

Spr'16



Inverse Modulo

Example: Find an inverse of 101 modulo 4620.

Solution:

$$4620 = 45 \cdot 101 + 75$$

$$101 = 1 \cdot 75 + 26$$

$$75 = 2 \cdot 26 + 23$$

$$26 = 1 \cdot 23 + 3$$

$$23 = 7 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$2 = 2 \cdot 1$$

$$1 = 3 - 1 \cdot 2$$

$$= 3 - 1 \cdot (23 - 7 \cdot 3)$$

$$= -1 \cdot 23 + 8 \cdot 3$$

$$= -1 \cdot 23 + 8 \cdot (26 - 1 \cdot 23)$$

$$= 8 \cdot 26 - 9 \cdot 23$$

$$= 8 \cdot 26 - 9 \cdot (75 - 2 \cdot 26)$$

$$= -9 \cdot 75 + 26 \cdot 26$$

$$= -9 \cdot 75 + 26 \cdot (101 - 1 \cdot 75)$$

$$= 26 \cdot 101 - 35 \cdot 75$$

$$= 26 \cdot 101 - 35 \cdot (4620 - 45 \cdot 101)$$

$$= -35 \cdot 4620 + 1601 \cdot 101.$$

1601 is an inverse of 101 modulo 4620



Congruence

- Developed by Gauss - a German mathematician
- Congruence is a statement about divisibility

If an integer $m \neq 0$ divides the difference $a-b$ we say that a is congruent to b modulo m and is written as $a \equiv b \pmod{m}$

- Theorem:

If $a \equiv b \pmod{m}$

&

$b \equiv c \pmod{m}$

then

$a \equiv c \pmod{m}$



Congruence

Application: Spr'23

- Using congruence we can find out whether the number is even or odd.
If $a \equiv 0 \pmod{2}$ then the number is even
And
If $a \equiv 1 \pmod{2}$ then the number is odd.
- To find out a day of a given particular date
- An integer n is divisible by 9 if and only if the sum of the digits of n is divisible by 9.

Linear Congruence

A congruence of the form

Spr'17,18

$$ax \equiv b \pmod{m},$$

where m is a positive integer, a and b are integers, and x is a variable, is called a **linear congruence**.



Linear Congruence

Solve the congruence $3x \equiv 4 \pmod{7}$

Solution:

$$7 = 2 \cdot 3 + 1$$

$$\Rightarrow 1 = 7 - 2 \cdot 3$$

Here, -2 is an inverse of 3 modulo 7 . Multiplying both sides of the congruence by -2 shows that

$$x \equiv 4 \cdot (-2) \pmod{7}$$

$$\equiv -8 \pmod{7}$$

$$\equiv 6 \pmod{7}$$

$$x \equiv 6 \pmod{7}$$

Linear Congruence

Solve the congruence $19x \equiv 4 \pmod{141}$

Solution:

$$141 = 7 \cdot 19 + 8$$

$$19 = 2 \cdot 8 + 3$$

$$8 = 2 \cdot 3 + 2$$

$$3 = 1 \cdot 2 + 1$$

$$1 = 3 - 1 \cdot 2$$

$$= 3 - 1 \cdot (8 - 2 \cdot 3)$$

$$= 3 - 8 + 2 \cdot 3 = 3 \cdot 3 - 8$$

$$= 3 \cdot (19 - 2 \cdot 8) - 8 = 3 \cdot 19 - 6 \cdot 8 - 8 = 3 \cdot 19 - 7 \cdot 8$$

$$= 3 \cdot 19 - 7(141 - 7 \cdot 19) = 3 \cdot 19 - 7 \cdot 141 + 49 \cdot 19 = 52 \cdot 19 - 7 \cdot 141$$

Here, 52 is an inverse of 19 modulo 141. Multiplying both sides of the congruence by 52 shows that

$$x \equiv 4 \cdot 52 \pmod{141} \equiv 208 \pmod{141} \equiv 67 \pmod{141}$$

Home Work: 10. Solve the congruence $4x \equiv 5 \pmod{7}$ Spr'18

11. Solve the congruence $7x \equiv 3 \pmod{15}$ Spr'23

Chinese Remainder Theorem

The Chinese Remainder Theorem was discovered by Chinese mathematician Sun Zi.

Spr'16,17

THE CHINESE REMAINDER THEOREM Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers greater than one and $a_1, a_2, \dots, a_n$ arbitrary integers. Then the system

$$x \equiv a_1 \pmod{m_1},$$

$$x \equiv a_2 \pmod{m_2},$$

.

.

.

$$x \equiv a_n \pmod{m_n}$$

has a unique solution modulo $m = m_1 m_2 \cdots m_n$. (That is, there is a solution x with $0 \leq x < m$, and all other solutions are congruent modulo m to this solution.)

Practical Problem:
[codeforces |](#)
[Remainders Game](#)

[hackerrank |](#)
[number-of-sequences](#)

Chinese Remainder Theorem

Example: Find the solution of the system using the Chinese remainder theorem.

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Solution: The solutions to this system are those x such that

$$x \equiv (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{m}$$

Given,

$a_1=2$	$m_1=3$	$\therefore m = m_1 \times m_2 \times m_3$	$M_1 = m/m_1 = 105/3 = 35$	$y_1 = M_1^{-1}$
$a_2=3$	$m_2=5$	$= 3 \times 5 \times 7$	$M_2 = m/m_2 = 105/5 = 21$	$y_2 = M_2^{-1}$
$a_3=2$	$m_3=7$	$= 105$	$M_3 = m/m_3 = 105/7 = 15$	$y_3 = M_3^{-1}$

We know,

$M_1 \times M_1^{-1} = 1 \pmod{m_1}$	$M_2 \times M_2^{-1} = 1 \pmod{m_2}$	$M_3 \times M_3^{-1} = 1 \pmod{m_3}$
$35 \times M_1^{-1} = 1 \pmod{3}$	$21 \times M_2^{-1} = 1 \pmod{5}$	$15 \times M_3^{-1} = 1 \pmod{7}$
$35 \times 2 = 1 \pmod{3}$	$21 \times 1 = 1 \pmod{5}$	$15 \times 1 = 1 \pmod{7}$
$\therefore y_1 = M_1^{-1} = 2$	$\therefore y_2 = M_2^{-1} = 1$	$\therefore y_3 = M_3^{-1} = 1$

$$\begin{aligned} \therefore x &\equiv (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{m} \\ &\equiv (2 \cdot 35 \cdot 2 + 3 \cdot 21 \cdot 1 + 2 \cdot 15 \cdot 1) \pmod{105} \\ &\equiv 233 \pmod{105} \\ &\equiv 23 \text{ (Answer)} \end{aligned}$$

Chinese Remainder Theorem

Example: Find the solution of the system using the Chinese remainder theorem.

Aut, Spr'16

$$x \equiv 2 \pmod{3}$$

$$x \equiv 4 \pmod{7}$$

$$x \equiv 6 \pmod{10}$$

Solution: The solutions to this system are those x such that

$$x \equiv (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{m}$$

Given,

$a_1=2$	$m_1=3$	$\therefore m = m_1 \times m_2 \times m_3$	$M_1 = m/m_1 = 210/3 = 70$	$y_1 = M_1^{-1}$
$a_2=4$	$m_2=7$	$= 3 \times 7 \times 10$	$M_2 = m/m_2 = 210/7 = 30$	$y_2 = M_2^{-1}$
$a_3=6$	$m_3=10$	$= 210$	$M_3 = m/m_3 = 210/10 = 21$	$y_3 = M_3^{-1}$

We know,

$M_1 \times M_1^{-1} = 1 \pmod{m_1}$	$M_2 \times M_2^{-1} = 1 \pmod{m_2}$	$M_3 \times M_3^{-1} = 1 \pmod{m_3}$
$70 \times M_1^{-1} = 1 \pmod{3}$	$30 \times M_2^{-1} = 1 \pmod{7}$	$21 \times M_3^{-1} = 1 \pmod{10}$
$70 \times 1 = 1 \pmod{3}$	$30 \times 4 = 1 \pmod{7}$	$21 \times 1 = 1 \pmod{10}$
$\therefore y_1 = M_1^{-1} = 1$	$\therefore y_2 = M_2^{-1} = 4$	$\therefore y_3 = M_3^{-1} = 1$

$$\therefore x \equiv (a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3) \pmod{m}$$

$$\equiv (2 \cdot 70 \cdot 1 + 4 \cdot 30 \cdot 4 + 6 \cdot 21 \cdot 1) \pmod{210}$$

$$\equiv 746 \pmod{210}$$

$$\equiv 116 \pmod{210} \text{ (Answer)}$$

Chinese Remainder Theorem

Example: Find x using Chinese remainder theorem, if possible, such that

Aut'22

$$2x \equiv 5 \pmod{7}, \text{ and}$$

$$3x \equiv 4 \pmod{8}$$

Solution:

Let, $2x \equiv 5 \pmod{7}$ ----- (1)

$$3x \equiv 4 \pmod{8} \text{ ----- (2)}$$

From (1),

$$2x \equiv 5 \pmod{7}$$

We know,

$$2 \times 2^{-1} = 1 \pmod{7}$$

$$\Rightarrow 2 \times 4 = 1 \pmod{7}$$

Multiplying (1) by 4,

$$x \equiv 20 \pmod{7}$$

$$\Rightarrow x \equiv 6 \pmod{7}$$

Let, $x \equiv 6 + 7a$ ----- (3)

From (2),

$$3x \equiv 4 \pmod{8}$$

$$\Rightarrow 3(6+7a) \equiv 4 \pmod{8} \quad [\text{from (3)}]$$

$$\Rightarrow 18+21a \equiv 4 \pmod{8}$$

$$\Rightarrow 2+5a \equiv 4 \pmod{8}$$

$$\Rightarrow 5a \equiv 2 \pmod{8} \text{ ----- (4)}$$

We know,

$$5 \times 5^{-1} = 1 \pmod{8}$$

$$\Rightarrow 5 \times 5 = 1 \pmod{8}$$

Multiplying (4) by 5,

$$a \equiv 10 \pmod{8}$$

$$\Rightarrow a \equiv 2 \pmod{8}$$

Let, $a \equiv 2 + 8b$ ----- (5)

From (3),

$$x \equiv 6 + 7a$$

$$\Rightarrow x \equiv 6 + 7(2+8b) \quad [\text{from (5)}]$$

$$\Rightarrow x \equiv 6 + 14 + 56b$$

$$\Rightarrow x \equiv 20 + 56b$$

$$\Rightarrow x \equiv 20 \pmod{56} \text{ (Answer)}$$



Chinese Remainder Theorem



Previous Question:

1. Find the value of x using the Chinese remainder theorem [**Spr'17**]

$$x \equiv 2 \pmod{6}, \quad x \equiv 3 \pmod{5}, \quad x \equiv 2 \pmod{2}$$

2. Find the value of x using the Chinese remainder theorem [**Spr'23**]

$$x \equiv 1 \pmod{2}, \quad x \equiv 2 \pmod{3}, \quad x \equiv 3 \pmod{5}$$

3. Find the value of x using the Chinese remainder theorem [**Aut'23**]

$$x \equiv 2 \pmod{3}, \quad x \equiv 1 \pmod{4}, \quad x \equiv 3 \pmod{5}$$